1. Standard Thick Client

A **thick client** (or fat client) is a full-featured PC that does most of its processing locally. It runs its own operating system and applications, and it can usually work even when the network is down. It uses the network mainly for file sharing, email, printing, and internet access.

- **Typical use:** office work such as Word, Excel, email, web browsing, and ERP or accounting software.
- **Typical specs:** a mid-range CPU (e.g., Core i5 or Ryzen 5), 8-16 GB RAM, a 256-512 GB SSD, and integrated graphics.
- **Pros:** works offline, gives good performance for the cost, and is flexible.
- **Cons:** each machine needs its own software installs, updates, and security patches, so it is harder to manage centrally than a thin client.

C*ontrast:* A **thin client** runs little locally and relies on a server to do the processing.

2. Graphics and CAD/CAM Design Workstation

Used for 2D/3D design, engineering drawings, and product modelling (AutoCAD, SolidWorks, CATIA, Blender). **CAD** is Computer-Aided Design and **CAM** is Computer-Aided Manufacturing, where the design drives machines like CNC mills.

- **CPU:** high clock speed, because many CAD operations are single-threaded.
- **GPU:** a professional graphics card (NVIDIA RTX A-series or AMD Radeon Pro) with certified drivers for stable rendering.
- **RAM:** 32-64 GB, with ECC memory preferred for reliability.
- **Storage:** a fast NVMe SSD for large project files.
- **Display:** one or more large, colour-accurate monitors (high resolution, wide colour gamut).

3. Audio/Video Editing Workstation

Used for editing video, mixing audio, motion graphics, and rendering (Premiere Pro, DaVinci Resolve, Pro Tools).

- **CPU:** many cores (8-16+), since rendering and encoding scale across cores.
- **GPU:** a strong GPU to accelerate effects, colour grading, and encoding.
- **RAM:** 32-128 GB, because 4K/8K footage is very memory-hungry.
- **Storage:** a tiered setup. A fast NVMe drive holds the OS, apps, and active projects, while large HDD or RAID arrays hold raw footage and archives.
- **Extras:** a colour-calibrated monitor, a low-latency audio interface, and good cooling for long renders.

4. Virtualisation Workstation

Runs several **virtual machines (VMs)** at once on a single physical computer, using tools like VMware Workstation, VirtualBox, Hyper-V, or Proxmox. It suits developers, testers, and IT staff who need to try different operating systems or simulate networks.

- **CPU:** many cores and threads, with hardware virtualisation support (Intel VT-x or AMD-V). Each VM needs its own share of cores.
- **RAM:** the most important resource, typically 32-128 GB or more, since each VM reserves its own memory.
- **Storage:** fast SSD/NVMe, because many VMs reading and writing at once creates heavy disk load.
- **Network:** multiple or fast NICs (e.g., 1-10 Gbps) if the VMs simulate real networks.
- **GPU:** usually not critical unless the VMs need GPU passthrough.